PRISTINE MONOLAYER $CrCl_3$ *First-Principles Electronic Structure & Magnetism*

Magnetic Moments by Chemical Species & Benchmark Tiers

Spin-resolved sphere projections (LORBIT = 11, Wigner-Seitz radii) across Pure PBE and Hubbard U tiers ($0 \le U \le 6$ eV)

TABLE 1: Atomic Site Magnetic Moments by Chemical Species (2×2 Supercell, $U_{Cr} = 3.29$ eV)

Chemical Species	Atoms / Cell (2×2)	Formal Valence & Configuration	Local Site Symmetry	m_s (μ_B)	m_p (μ_B)	m_d (μ_B)	m_{tot} / Atom (μ_B)	Sublattice Total (μ_B)
Cr (Chromium)	8	Cr^{3+} ($3d^3, t_{2g}^3 e_g^0$)	Octahedral (D_{3d})	+0.031	+0.039	+3.047	**+3.117**	**+24.936**
Cl (Chlorine)	24	Cl^- ($3s^2 3p^6$)	Bridging (C_{2v})	-0.001	-0.063	0.000	**-0.064**	**-1.536**
Interstitial / Delocalized	--	Delocalized spin density	Inter-atomic space	--	--	--	**--**	**+0.600**
Unit Cell Total (1×1)	**2 Cr + 6 Cl**	**Formula unit: $CrCl_3$**	**Honeycomb (P-31m)**	**+0.060**	**-0.300**	**+6.094**	**+3.000 / f.u.**	**+6.000**
Supercell Total (2×2)	**8 Cr + 24 Cl**	**8 formula units (Cr_8Cl_{24})**	**Supercell (P-31m)**	**+0.216**	**-1.196**	**+24.372**	**+3.000 / f.u.**	**+24.000**

TABLE 2: Methodological Benchmark Across Hubbard U_{Cr} Parameter Sweep ($0 \le U \le 6$ eV)

Methodological Functional Tier	Hubbard U_{Cr} (eV)	Cr-$3d$ (m_d) (μ_B)	Cr Total (m_{Cr}) (μ_B)	Cl-$3p$ (m_p) (μ_B)	Cl Total (m_{Cl}) (μ_B)	M_{tot} / Cell (μ_B)	Band Gap E_g (eV)	Physical Description & Electronic Status
Pure PBE Baseline	0.00	+2.841	+2.908	-0.022	-0.022	3.00 / f.u.	1.50 (dir)	Underestimated gap, high covalency
PBE+D3 (No U)	0.00	+2.841	+2.908	-0.022	-0.022	3.00 / f.u.	1.50 (dir)	vdW included; identical local moments
PBE+D3+U_{Cr}	1.00	+2.907	+2.975	-0.035	-0.036	3.00 / f.u.	1.82 (dir)	Initial d-electron localization
PBE+D3+U_{Cr}	2.00	+2.971	+3.040	-0.048	-0.049	3.00 / f.u.	2.18 (dir)	Progressive gap widening
PBE+D3+U_{Cr}	3.00	+3.033	+3.103	-0.060	-0.062	3.00 / f.u.	2.48 (dir)	Approaching optimal lattice compliance
PBE+D3+U_{Cr} ★ Optimal	**3.29**	**+3.047**	**+3.117**	**-0.063**	**-0.064**	**3.00 / f.u.**	**2.56 (dir)**	**Matches experimental $a_0 = 6.056$ Å & LR**
PBE+D3+U_{Cr}	4.00	+3.094	+3.165	-0.073	-0.075	3.00 / f.u.	2.59 (dir)	Saturating gap opening
PBE+D3+U_{Cr}	5.00	+3.153	+3.226	-0.086	-0.088	3.00 / f.u.	2.68 (dir)	Strongly localized limit
PBE+D3+U_{Cr}	6.00	+3.211	+3.285	-0.098	-0.100	3.00 / f.u.	2.77 (dir)	Upper correlation bound

① Cr^{3+} Local Moment ($3d^3$)

- Octahedral ligand field (D_{3d}) splits $3d$ levels into an occupied t_{2g}^3(↑ ↑ ↑) triplet and empty e_g^0 doublet.
- High-spin $S = 3/2$ produces $3.000\,\mu_B$ formal spin.
- Projected Wigner-Seitz moment: $+3.117\,\mu_B$ (98% localized in $3d$ manifold: $m_d = 3.047\,\mu_B$).

② Cl^- Negative Spin Polarization

- Ligand Cl^- ($3p^6$) exhibits net negative polarization opposite to Cr: $m_{tot}(Cl) = -0.064\,\mu_B$.
- Arises from Goodenough–Kanamori superexchange and covalent virtual hopping between Cr-$3d$ and Cl-$3p$.
- High symmetry: all 24 Cl atoms in 2×2 are equivalent and carry identical negative polarization.

③ Integer Integer Total Magnetization

- Total cell moment is strictly integer: $M_{tot} = 24.000\,\mu_B$ (2×2, 8 f.u.) = $3.000\,\mu_B$/f. u.
- Ferromagnetic insulator ground state with integer valence band filling and clean gap to empty states.
- Interstitial space carries $+0.60\,\mu_B$ background balancing local Cr core and Cl ligand projections.